

ECE 1387 Lecture #6

Oct 17,
2025

- Analytical placement
- Simulated annealing-based placement

Finding $P(b_i)$ Set of paths for
overfilled bin b_i

Use BFS (breadth-first search)

demand = 0

mark all bins unvisited

visited (b_i) = true

insert b_i into an empty path P

add path P to FIFO Q , Q

repeat

pop P from Q

tail bin = current end bin of P

for neighbour bins b_k of tail bin

if (visited (b_k))

continue

cost = compute cost (tail bin, b_k)

if cost $< \infty$

Create a copy of P , P_{copy}

cost = cost(P_{copy}) + cost

insert b_k into p_{copy} // new tail
visited(b_k) = true

if (b_k is empty) then
 insert p_{copy} in $P(b_i)$
 demand ++;

else
 add p_{copy} to Q

end if

end for

until Q is empty or demand $\geq$ supply(b_i)

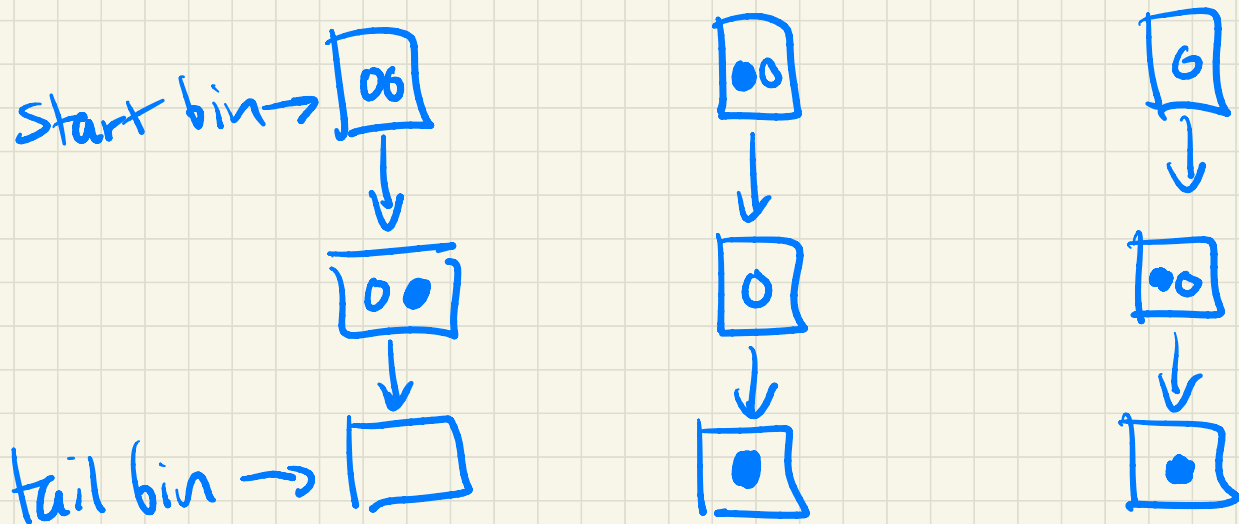
Cost $\propto \psi$ (tailbin, b_k)

→ find a cell in tailbin that can move to b_k s.t. quadratic distance from the cell's solved placement and new position in b_k is minimized and less than ψ

If no such cell exists, Cost = ∞

Otherwise Cost = quadratic displacement of cell from orig. solved placement.

Moving cells along a path P_k
(from tail bin towards start bin)

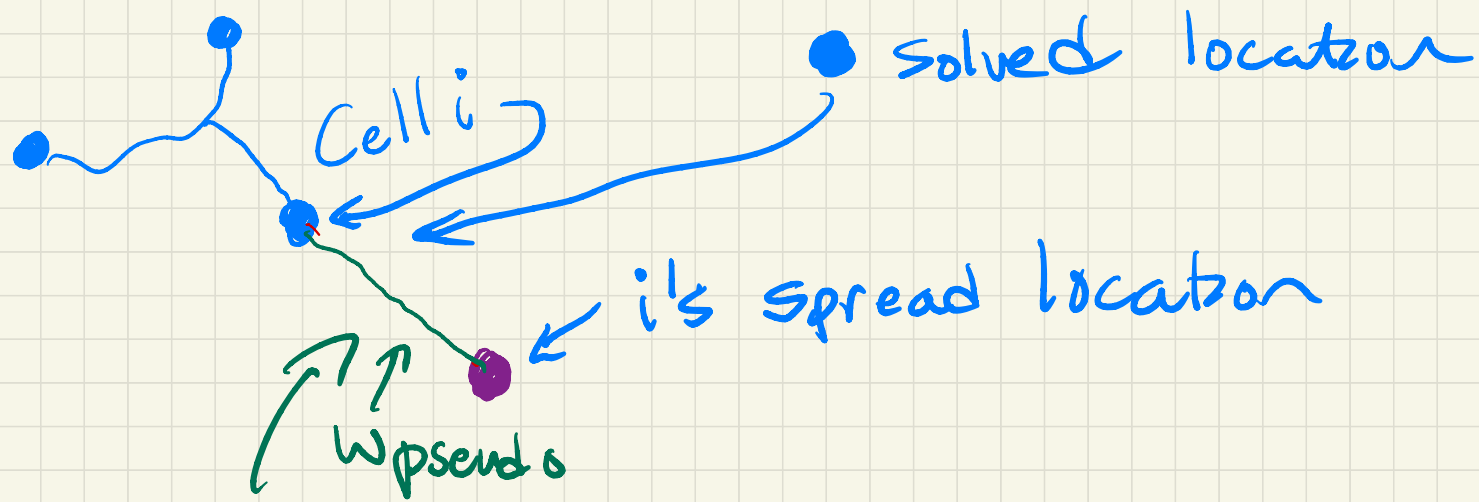


- check costs again to not exceed ψ . (because of overlapping paths)

After spreading, we should have
a placement with no
overused bins! $\rightarrow$ great

If we solve AP system again,
get same solⁿ back! bad

$\rightarrow$ Want to modify AP system so
cells gradually move towards
spread locations.



- Create a fake (pseudo) net between i and its spread location
- Weight of pseudo connection is increased with each AP iteration

In essence, we create an
"anchor" at spread position
with a connection to the anchor.

How to bring cells towards
their spread position?

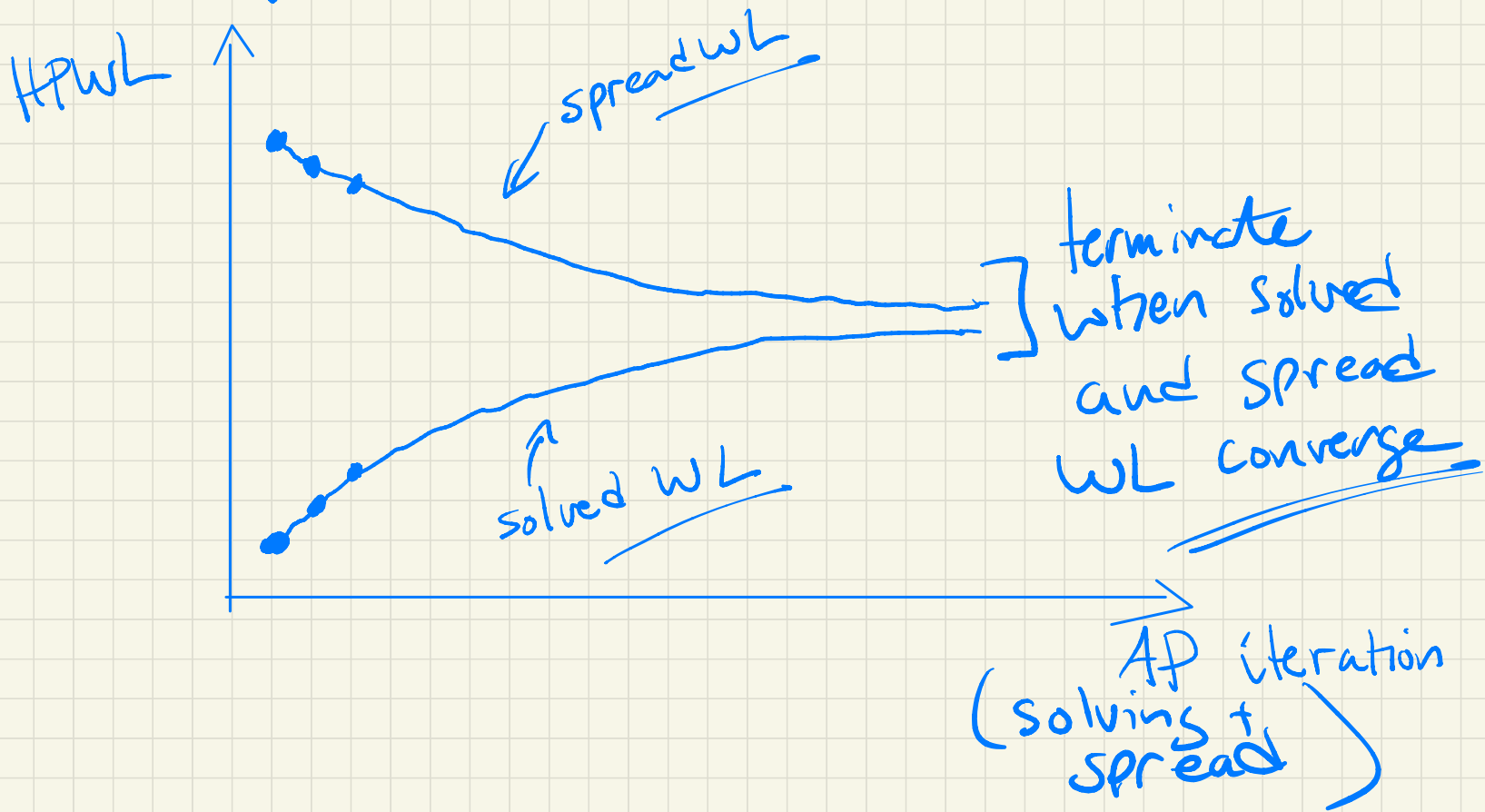
Ans: Pseudo (fake) nets + pseudo weights

Which part of math does it affect?

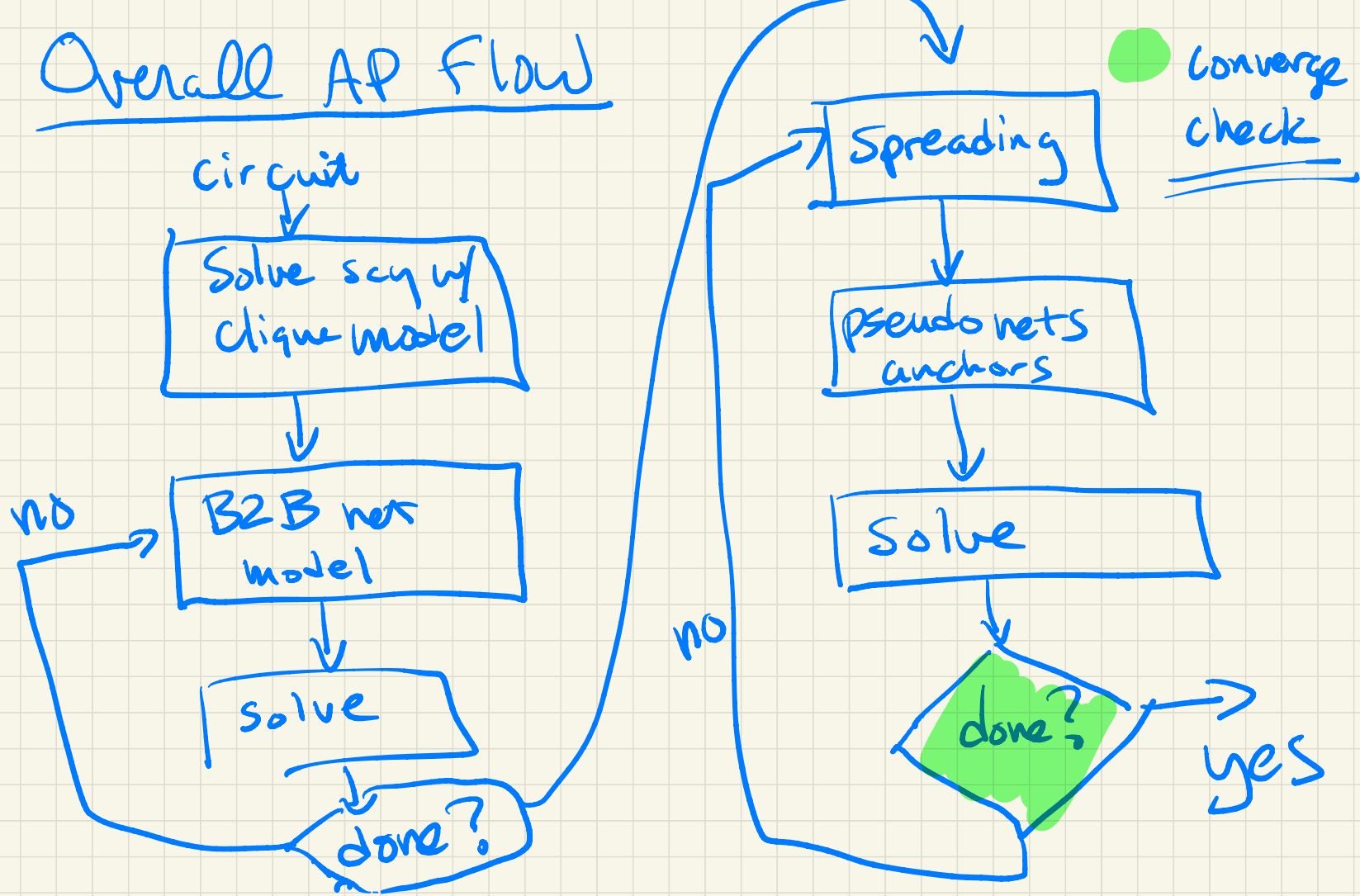
- right hand side, $\vec{b}_x$
 $\vec{b}_y$

- on-diag elements of Q_x, Q_y

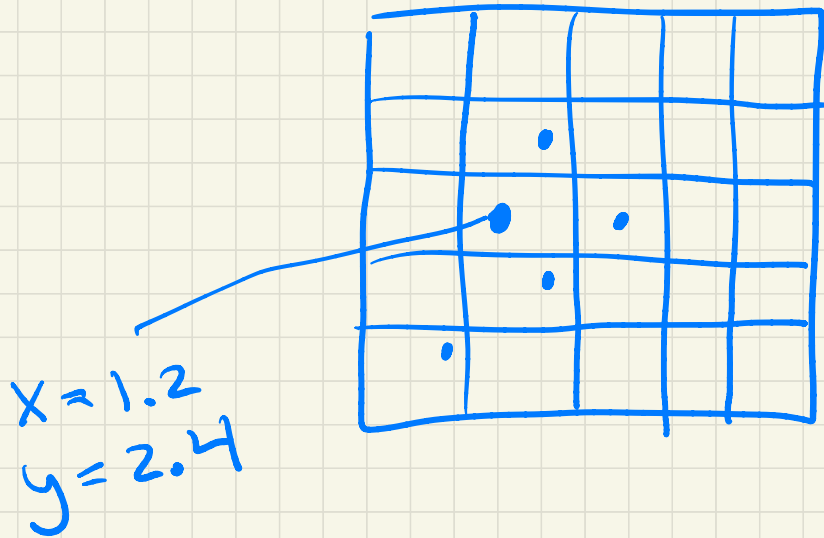
Convergence



Overall AP Flow



AP "fitting" find legalization



One way: find legal grid point closet to solved location.

→ Snap cell to that point if its vacant.

if closet slot is not vacant,
do BFS to find closet
vacant slot.

Simulated Annealing-Based Placement

- SA is a very general optimization strategy
- Can be applied to almost any problem
- IBM Research 1980s
- Based on an analogy
"annealing" of metals

- to make strong steel, melt and cool slowly
- atoms coalesce into a regular crystal (hard to break)

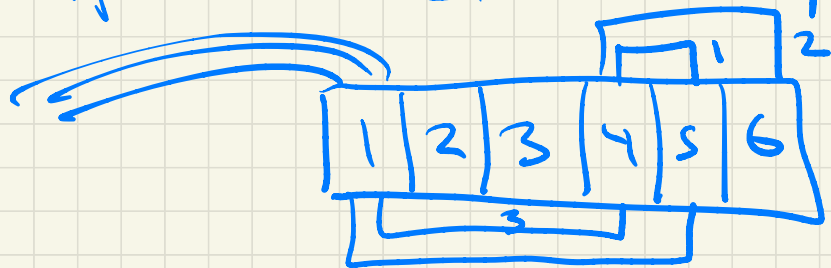
SA has a "temperature" notion

T : algorithm parameter

High T : e.g. in placement cells
move cell over the chip

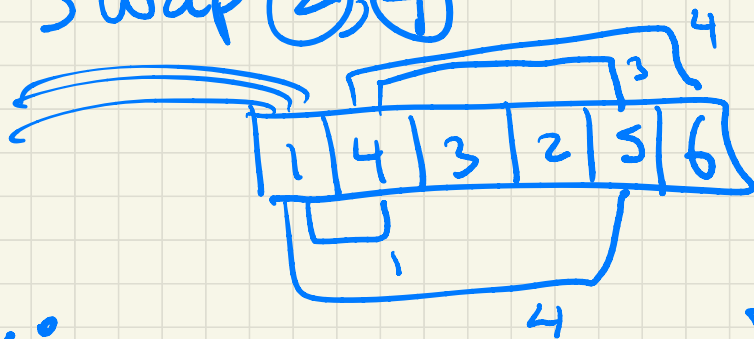
Low T : restricted cell movements

Important SA Concept "Hill Climbing"



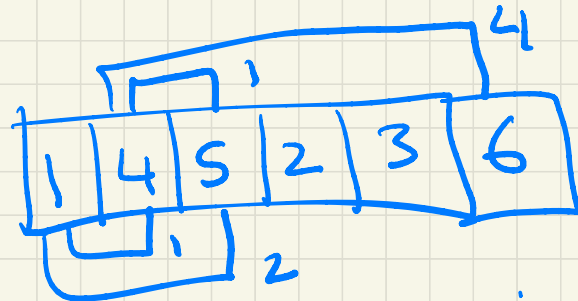
WL = 10

Swap (2), (4)⁴



WL = 12 (worse)

now... Swap (5), (3)



WL = 8

Hill climbing allows a "move"
(a perturbation) where cost
gets worse... in hope of finding
a better cost later.

As opposed to a greedy algorithm
which always demands cost
be reduced.

General Framework of SA

- ① Need an initial constructed state, j_0
e.g. HPWL
- ② Need a cost function, $\text{cost}(\text{state})$
- ③ Need an initial (high) temperature T_0

$SA(j_0, T_0) \{$ in generic form

$$T = T_0$$

$$X = j_0 \quad // \quad X = \text{current state}$$

while ("stopping criterion not met") {

while ("inner loop criterion not met") {

$$j = \text{generate}(X) \quad // \quad j_{\text{new state}}$$

$$\text{calculate } \Delta C = // \text{cost change} \\ \text{cost}(j) - \text{cost}(X)$$

all @
one temperature

if (Accept($\Delta c, T$))

$X = j$ // accept "move"

} // inner loop

$T = \text{update}(T)$

} // outer loop

}